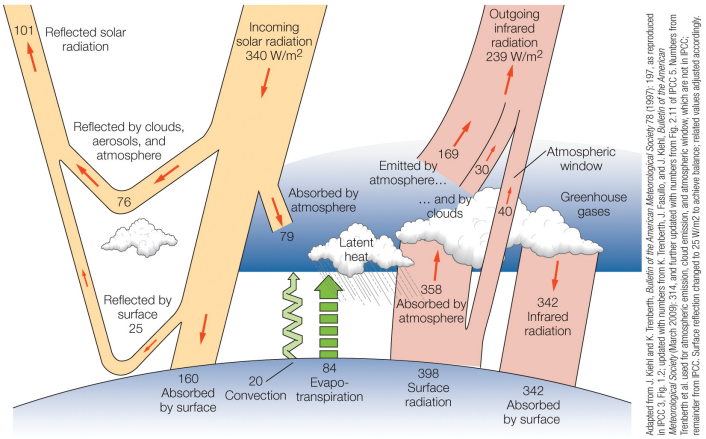
Physics 12 Winter 2026
Week 10 Discussion

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1. **Summary of Topics from Last Week**

   - *From the lectures*
     - ◇ Revisited: radiation, insolation
     - ◇ Radiation energy balance equation $\frac{S}{4} = e\sigma T^4$
     - ◇ Effect of Earth's atmosphere (greenhouse effect)
       - ∗ Acts as a blanket for the Earth.
       - ∗ Raises Earth's surface temperature by 33 K to 288 K.
       - ∗ The trapped heat will still escape ultimately, but the "blanket" slows down the escape for infrared radiation.
     - ◇ Mechanism of greenhouse effect on the molecular level
     - ◇ Zero-dimensional energy balance model
       - ∗ Treats the Earth as a point
       - ∗ $S = 1360$ W/m^2 is the solar constant at the Earth's location.
       - ∗ Usually assume emissivity $e = 1$ (perfect blackbody)
       - ∗ Also have to consider albedo (reflection)
     - ◇ Selective absorption, thermostatic effect (negative feedback)
       - ∗ Hotter (colder) atmosphere emits shorter (longer) wavelength
       - ∗ Shorter (longer) wavelength radiation is less (more) trapped by greenhouse gases and escapes to space (gets re-emitted onto the surface)
       - ∗ Atmosphere temperature drops (rises)
     - ◇ Comparison of Earth, Venus and Mars
       - ∗ Distance to the sun has only moderate effect on surface temperature.
       - ∗ Venus has runaway greenhouse effect due to dense atmosphere with mostly CO_2.
       - ∗ Mars has very little greenhouse effect due to very thin atmosphere.
     - ◇ Detailed description of energy flow on Earth (Sun, surface, atmosphere)



- ◇ Stratospheric cooling
  - ∗ Greenhouse gases heat up the atmosphere by giving off energy to other air molecules through radiation/collisions.
  - ∗ They can't do that efficiently in the less dense stratosphere–more energy lost to space through radiation.
  - ∗ This effect is a fingerprint of global warming by greenhouse gases.
- ◇ Effect from albedo, convection, evapotranspiration
- ◇ Climate forcing, sensitivity
  - ∗ Climate forcing: imposed imbalance (in units of W/m^2), equal to incoming (absorbed) solar radiation minus the outgoing infrared radiation.
  - ∗ Climate sensitivity: change in Earth's surface temperature in response to a change in climate forcing (in units of $°C/(W/m^2)$)
- ◇ Feedback (positive and negative)
  - ∗ See examples in slides and homework problems.
- ◇ Natural and anthropogenic forcing
  - ∗ Natural: volcanism, solar fluctuation, Earth's orbital changes
  - ∗ Anthropogenic: greenhouse gases (+ forcing), ozone (also a GHG), aerosols (mostly − forcing), albedo change due to land use
- ◇ Global warming potential, atmospheric lifetime

- *From the textbook* (Wolfson 4th edition)
  - ◇ Energy balance equation, Greenhouse effect (Chapter 12.1-3)
  - ◇ Earth's energy balance and imbalance, comparison with other planets (Chapter 12.4-5)
  - ◇ Climate forcing, climate sensitivity, feedback effects (Chapter 13.1-3)
  - ◇ Natural and anthropogenic forcings, carbon cycle (Chapter 13.4-5)

2. **Brain-warmer/mock quiz.**

(a) **Climate forcing.** The best estimate for the climate forcing due to changes in the Sun's activity is 0.01 W/m^2 (according to the textbook). Estimates put Earth's climate sensitivity at around 0.5-1.2 $K/(W/m^2)$ (according to ChatGPT...). Given this range, what would be the range of temperature increase we get due to changes in solar activity?

A. 0.5-1.2°C
B. 5-12°C
C. 0.005-0.012°C
D. 0.01-0.024°C
E. 0.05-0.12°C

Answer: C. To find the increase in temperature due to solar activity, multiply the climate forcing due to solar activities with the Earth's climate sensitivity.

(b) **Stratospheric cooling.** Which of the following statements are true about stratospheric cooling? (Select all that apply)

A. Stratospheric cooling occurs when the troposphere cools as well.
B. Stratospheric cooling is a fingerprint of global warming by a greenhouse gas.
C. Stratospheric cooling is the result of a negative climate forcing due to the accumulation of particulates that originate from volcanism.
D. Stratospheric cooling occurs because the low air density in the stratosphere results in less efficient transfer of absorbed heat from CO_2 to other air molecules.

Answer: B and D. Both statements can be found in Box 12.1 in the textbook.

(c) **Global warming potential.** Which of the following statements are true about the global warming potential (GWP) of a greenhouse gas? (Select all that apply)

A. CO_2 has a GWP of 1.
B. The only factor that determines the GWP of a greenhouse gas is its atmospheric lifetime.
C. The GWP for a greenhouse gas depends on the timescale of interest.
D. Nitrous oxide (N_2O) has a shorter atmospheric lifetime than methane (CH_4), and therefore a lower GWP.

E. The GWP for a greenhouse gas can be measured either on a per-mass basis or on a per-molecule basis.

Answer: A, C and E. A is by definition. GWP is determined by many factors, and its significance depends on the way it is measured, which should be specified. Information about the GWP for common greenhouse gases can be found in Table 13.1.

(d) **Earth's twin.** Which of the following statements are true about Venus? (Select all that apply)

A. Venus's atmosphere is more than 90% CO_2.

B. Venus's atmosphere is less dense (lower pressure) than that of the Earth.

C. Venus's surface temperature is much higher than that of the Earth.

D. Venus is more reflective of sunlight than Earth.

E. Without the greenhouse effect, Venus would still have a slightly higher surface temperature than the Earth due to its proximity to the Sun.

Answer: A, C and D. Venus's atmosphere is 96% CO_2 and has a pressure that's almost 100 times higher than that of the Earth. Its surface temperature is 462°C (735 K) compared to Earth's 15°C (288 K). It has a higher albedo (0.75) than the Earth (0.30), meaning it's more reflective. Accounting for this higher reflectivity but ignoring the greenhouse effect, its surface temperature would have been lower than that of the Earth despite its proximity to the Sun (see Table 12.1).

(e) **Earth's atmospheric blanket.** Which of the following statements are true about the effect of the Earth's atmosphere? (Select all that apply)

A. Looking from the outside, the Earth-atmosphere system behaves like a blackbody that has the same temperature as the surface temperature of an identical twin of Earth that has no atmosphere.

B. Due to the greenhouse effect on the Earth, the surface temperature of the Earth is raised by 18°C.

C. The average surface temperature of the Earth is about 15°C (288 K).

D. The ozone layer is mainly in the upper stratosphere.

Answer: A and C. Without the atmosphere, the Earth would have a surface temperature of −18°C (255 K). Since the atmosphere acts like a blanket and only traps the heat inside itself, the Earth-atmosphere system would have a similar blackbody temperature. Only the surface inside would have a higher temperature. The temperature increase due to the greenhouse effect on Earth is 33°C, resulting in an average surface temperature of 15°C (288 K). The ozone layer is mainly in the lower stratosphere.

3. **Effect of Albedo**[1]**.** Use a simple zero-dimensional energy-balance model without atmosphere to estimate Venus's average temperature, (a) ignoring reflection and (b) assuming an albedo of 0.75. Venus is 108 million kilometers from the Sun, versus Earth's 150 million kilometers.

   Hint: Because sunlight intensity falls as the inverse square of the distance, the solar constant at Venus is a factor of $(150/108)^2$ larger than at Earth.

   Solution:

   (a) $S(r) = \text{constant}/r^2$; therefore,

$$\frac{S_{\text{Venus}}}{S_{\text{Earth}}} = \frac{\text{constant}/r_{\text{Venus}}^2}{\text{constant}/r_{\text{Earth}}^2} = \frac{r_{\text{Earth}}^2}{r_{\text{Venus}}^2} = \left(\frac{150}{108}\right)^2 \tag{1}$$

   Sure enough, the solar constant at Venus, S_{Venus}, is a factor $(150/108)^2$ larger than the solar constant at Earth, S_{Earth}:

$$S_{\text{Venus}} = S_{\text{Earth}} \left(\frac{150}{108}\right)^2 = (1360\ \text{W/m}^2) \left(\frac{150}{108}\right)^2 = 2623\ \text{W/m}^2 \tag{2}$$

   Neglecting reflection, the temperature is

$$T = \left(\frac{S_{\text{Venus}}}{4\sigma}\right)^{1/4} = \left(\frac{2623\ \text{W/m}^2}{4 \cdot 5.67 \times 10^{-8}\ \text{W/m}^2/\text{K}^4}\right)^{1/4} = 328\ \text{K}. \tag{3}$$

   (b) Accounting for the albedo $\alpha = 0.75$ means Venus absorbs only the fraction $1 - \alpha$ of the incident sunlight:

$$S'_{\text{Venus}} = (1 - \alpha) S_{\text{Venus}} = (0.25)(2623\ \text{W/m}^2) = 656\ \text{W/m}^2 \tag{4}$$

   so the temperature with reflection is

$$T = \left(\frac{S'_{\text{Venus}}}{4\sigma}\right)^{1/4} = \left(\frac{656\ \text{W/m}^2}{4 \cdot 5.67 \times 10^{-8}\ \text{W/m}^2/\text{K}^4}\right)^{1/4} = 232\ \text{K}. \tag{5}$$

[1] Wolfson (4th edition) Chapter 12 Exercise 3